

Toshio Shinoda^a
Toru Aizawa^b
Toshihide Shiota^b
Masafumi Katakura^b
Takashi Yamada^b
Hideo Arakura^a
Yu Tai Chang^c
Shin-ya Kobayashi^c
Tetsutaro Mizukami^d

^a Department of Artificial Organ Treatment,

^b Department of Geriatrics,
Endocrinology and Metabolism and

^c Second Department of Surgery, Shinshu
University School of Medicine, Matsumoto;

^d Kashiwabara Clinic, Kashiwabara-Kanbara,
Minamiazumi-gun, Japan

Exacerbation of Latent Heart Failure by Mild Hypocalcemia after Parathyroidectomy in a Long-Term Hemodialysis Patient

Key Words

Hypocalcemia

Heart failure

Hemodialysis

Latent myocardial dysfunction

Parathyroidectomy

Abstract

A 60-year-old uremic patient treated with hemodialysis for 13 years developed acute hypotension during hemodialysis and overt heart failure subsequently in a period of relatively mild hypocalcemia after parathyroidectomy. Blood pressure during hemodialysis was maintained by continuous calcium infusion alone. Cardiomegaly was improved by normalization of serum calcium. The cardiac complication is considered to be exacerbation of the underlying myocardial dysfunction by the acute decline of serum calcium which might be a feature of the 'hungry bone' syndrome. Latent myocardial dysfunction might be exacerbated by only mild hypocalcemia after parathyroidectomy in long-term hemodialysis patients.

Introduction

The serum calcium level is of fundamental importance in the cardiac contraction process [1]. Hemodialysis reportedly improves left ventricular function not only by a Frank-Starling effect due to volume reduction but also by an increase in the cardiac contractility possibly due to an increase in serum calcium [2]. Although hypocalcemia is a common clinical feature of chronic renal failure, it rarely causes an overt depression of cardiac contractility in uremic

patients. To the best of our knowledge, only 2 uremic patients who developed heart failure as a result of hypocalcemia have been reported [3, 4]. Here we report a uremic patient treated with hemodialysis for 13 years who developed acute hypotension during hemodialysis and overt heart failure subsequently by relatively mild hypocalcemia after parathyroidectomy. Possible mechanisms to develop hypocalcemic heart failure are suggested from a consideration of the patient and the cases in the literature.

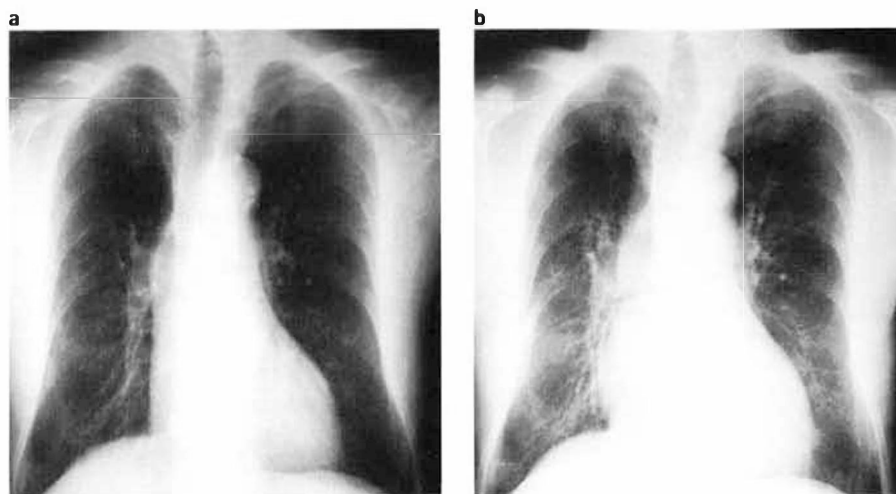


Fig. 1. X-ray films of the chest. **a** Before operation. CTR is 49% with no pulmonary congestion. Deformity of the thorax and calcification of the trachea and aortic knob are observed. **b** The 8th POD. CTR was 54% with moderate pulmonary congestion.

Case Report

A 60-year-old man was admitted to the hospital because of renal osteodystrophy.

He had been well until 15 years before, when a diagnosis of chronic glomerulonephritis was made. He had been treated with hemodialysis at private clinics for 13 years. Nine years before entry, cardiac arrest had occurred several times during hemodialysis. In subsequent years, blood pressure was stable but low normal during hemodialysis, and occasional ventricular ectopic beats were observed. Ubidecarenone and disopyramide had been administered in the past 2 years. The cardiothoracic ratio (CTR) on the chest X-rays had been mostly less than 50% in those 5 years.

Fifteen months before entry, a diagnosis of renal osteodystrophy due to secondary hyperparathyroidism was made, and parathyroidectomy was indicated. Deferoxamine 500 mg a week had been infused for 9 months before parathyroidectomy for the treatment of aluminum deposition in the bones which was diagnosed by the deferoxamine infusion test.

Blood pressure was 118/74 mm Hg, the pulse 85 bpm and respiration 20 breaths/min. Generalized skin pigmentation, anemia and a grade 2 systolic heart murmur in the apical region were observed. Physical findings were otherwise normal.

Serum chemistry data on admission included: urea nitrogen 106 mg/100 ml, creatinine 13.4, calcium 10.1 and phosphorus 5.5 mg/100 ml, total protein 6.4 g (albumin 3.1 and globulin 3.3 g) per 100 ml, alkaline phosphatase 435 U/l (normal 29–98), sodium 141 and potassium 5.4 mmol. Hemoglobin level was 9.4 g/100 ml. C-terminal parathyroid hormone level was 12.8 ng/ml, free thyroxine level 0.51 ng/dl and TSH 2.8 μ IU/ml (normal range: <0.35, 0.8–2.0 and 0.2–4.0, respectively).

An electrocardiogram (EKG) demonstrated a sinus rhythm at a rate of 73 with the P–R interval of 0.25 s and the corrected Q–T interval (Q–T_c) 0.44 s; specific S–T-segment and T-wave abnormalities were not observed. X-ray films of the chest showed no cardiac enlargement (CTR 49%) or pulmonary congestion (fig. 1a). An echocardiographic examination revealed a substantial decrease of the left ventricular contractility.

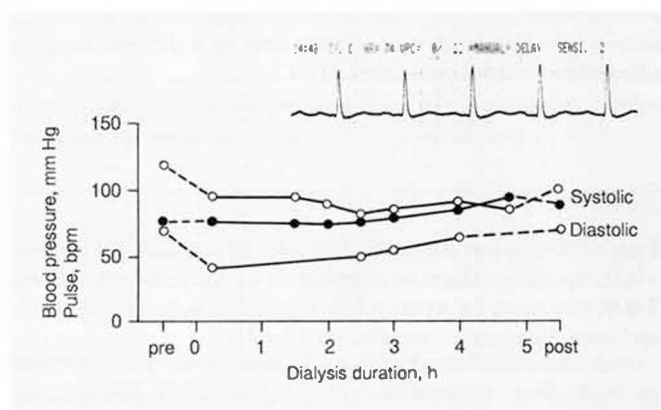


Fig. 2. Blood pressure (○), pulse (●) and ECG monitoring during hemodialysis in the preoperative course (the 6th hospital day). Blood pressure was rather stable despite persistent hypotension. Total fluid removal was 1,000 ml. Serum ionized calcium level was 1.38 mM immediately after initiation of hemodialysis.

On the 11th hospital day, total parathyroidectomy with autotransplantation was performed. The parathyroid glands weighed 4.7 g as a total, showing diffuse chief cell hyperplasia histologically. In the preoperative course, the blood pressure was low normal but stable during hemodialysis (fig. 2).

On the first postoperative day (POD), serum calcium was 7.8 mg/100 ml with an infusion of calcium gluconate. Oral administration of calcium lactate 6 g and calcitriol 3 μ g per day was begun on the 2nd POD. Acute hypotension as low as 60 mm Hg and dyspnea developed 3 h after the initiation of hemodialysis on that day (fig. 3). Tetany did not precede this episode. As these complications were refractory to massive saline infusion, hemodialysis was interrupted. The fluid remo-

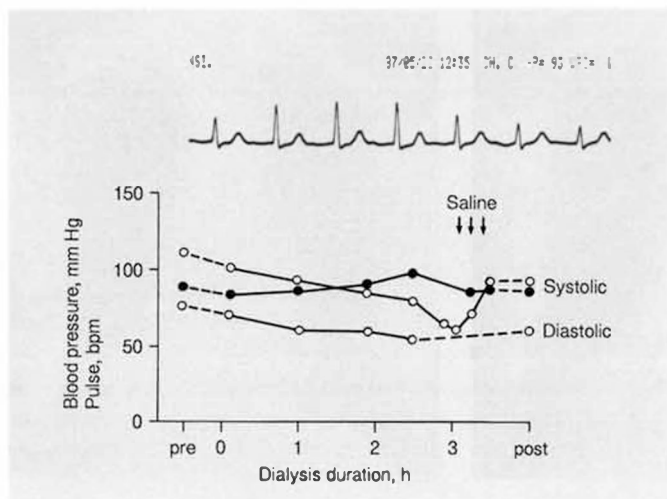


Fig. 3. Blood pressure (○), pulse (●) and ECG monitoring during hemodialysis on the 2nd POD. The systolic pressure decreased to as low as 60 mm Hg 3 h after initiation of hemodialysis. Despite a total of 700 ml saline infusion, it was impossible to continue hemodialysis with fluid removal. Hemodialysis was interrupted 3.5 h after the initiation and total fluid removal was only 600 ml.

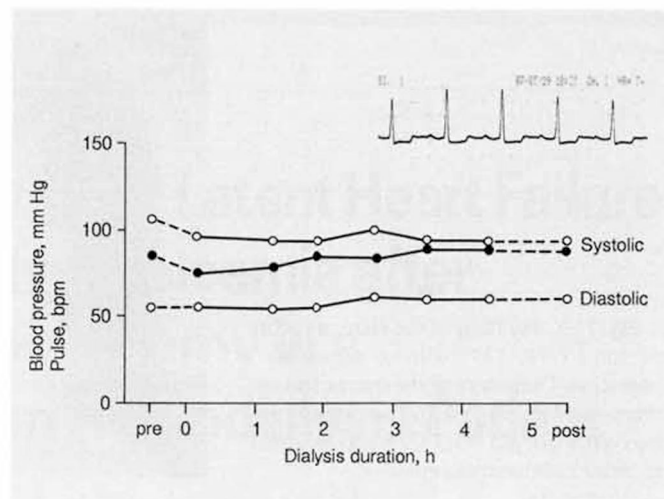


Fig. 4. Blood pressure (○), pulse (●) and ECG monitoring during hemodialysis on the 9th POD. The hemodynamic parameters were stable throughout the procedure with continuous calcium infusion, although 1st degree atrioventricular block was observed on the ECG monitoring.

val rate on the day was 400 ml/h which rate was comparable to that on the 6th hospital day. Dialysate composition was the following: sodium 135 mM, potassium 2.5, calcium 1.75, magnesium 0.75, chloride 106.5, bicarbonate 30, acetate 8 and glucose 8.3 mM.

On the 3rd POD, hemodialysis was carried out for 3 h without fluid removal because of hypotension. On the 5th POD, dyspnea and tachycardiac atrial fibrillation were present prior to hemodialysis. Sequential ultrafiltration and hemodialysis was carried out to remove excessive body fluid. A total of 2,100 ml fluid was removed, but acute hypotension as low as 60 mm Hg systolic and multiple ventricular ectopic beats developed during the procedure despite the administration of pressor drugs and antiarrhythmics.

Predialysis serum calcium level was not severely low, but substantially lower than the preoperative values and the increase in serum calcium by hemodialysis was insufficient in this postoperative period (table 1). EKG on the 2nd POD showed a slight prolongation of Q-T (0.45 s). Because of questionable hypocalcemic hypotension, we initiated a continuous infusion of calcium gluconate 2 mEq/h during hemodialysis without changing the oral dose of calcium and vitamin D.

This procedure resulted in a substantial increase in serum calcium after hemodialysis (table 1) and disappearance of acute hypotension and arrhythmias during hemodialysis (fig. 4). Predialysis serum calcium level was not appreciably changed by this procedure. The change in serum potassium level during hemodialysis with continuous calcium infusion was comparable to that in the case without calcium infusion (table 1). Although cardiomegaly and pulmonary congestion did not improve immediately after calcium infusion (fig. 1b), they disappeared as the serum calcium level normalized in 4 months after parathyroidectomy.

Table 1. Serum electrolytes and arterial blood pH

	2nd POD		3rd POD		5th POD		9th POD ^a	
	pre	post	pre	post	pre	post	pre	post
Ionized calcium mM	1.01	1.10	0.95	1.04	1.03	1.16		
Total calcium mg/100 ml	6.9	7.2			7.3		7.3	8.7
Potassium, mM	5.4	4.5					5.1	4.5
Arterial blood pH	7.40	7.52	7.43	7.50	7.40	7.48		

^a With calcium infusion.

Discussion

We consider that latent heart failure was exacerbated by hypocalcemia during the postoperative course in our patient. The initial sign of cardiac dysfunction was acute hypotension during hemodialysis, and the consequence

Table 2. Cases with heart failure due to hypocalcemia

Case No.	Age Sex years	Etiology of hypocalcemia	Serum calcium at onset mg/100 ml	Preceding duration of hypocalcemia	Reference No.
1	45 F	TX	5.8	16 years?	10
2	43 M	IHP	4.1	6 months?	11
3	58 F	IHP	3.7	40 years?	12
4	38 F	IHP	4.9	?	13
5	12 F	IHP	3.9	?	14
6	41 ?	TX	7.4	?	15
7	9 M	IHP	4.6	?	16
8	15 M	rickets	7.2	?	17
9	11 F	uremia	5.0	several days	3
10	14 F	IHP	3.5	4 years?	18
11	19 M	PTX	6.0–6.6	6 days	5
12	35 M	MHS	4.0	?	19
13	10 F	IHP	3.2	?	20
14	35 M	IHP	4.1	6 years?	20
15	47 F	TX	5.6	5 years?	8
16	76 F	PTX	5.1	9 months?	21
17	39 F	TX	6.2	10 years	22
18	61 M	IHP	6.0 (6.2 ^a)	?(6 months ^a)	23
19	37 M	uremia	3.9	?	4

TX = Thyroidectomy; IHP = idiopathic hypoparathyroidism; PTX = parathyroidectomy; MHS = Monilia-hypoparathyroidism syndrome.

^a Second admission.

was overt heart failure. The following 2 reasons strongly supported this hypothesis. First, the hypotensive episodes during hemodialysis were refractory to a massive saline infusion and pressor drugs, but were promptly improved by a continuous calcium infusion alone. Second, cardiac enlargement during the hypocalcemic period was improved as serum calcium level normalized.

Despite the fundamental importance of calcium in the cardiac contraction process, chronic hypocalcemia has rarely been shown to result in heart failure [1]. A small number of cases (excluding infants) with hypotension or heart failure due to hypocalcemia have been reported (table 2). One of the reasons for the rarity of hypocalcemic heart failure is that the presence of tetany allows physicians to discover and correct hypocalcemia before the development of hypotension or heart failure [5]. It is also suggested that hypocalcemia causes heart failure only in the presence of underlying organic heart disease, pulmonary embolism, or fluid overload [5]. Existence of pronounced hypocalcemia for a long period is considered to be a prerequisite for the development of heart failure [6].

In this sense, uremic patients are considered to be candidates for the development of hypocalcemic heart failure, since chronic renal failure often accompanies chronic hypocalcemia, volume overload and myocardial injury. Nevertheless, only 2 uremic patients with hypocalcemic heart failure have been reported: 1 with pericarditis [3] and another with severe hypocalcemia [4]. Considering the present case together with the previously reported cases, we think that an acute decline in serum calcium is an additional prerequisite for the development of hypocalcemic heart failure.

Acute hypocalcemia reportedly depresses ventricular function. The postoperative absolute calcium level was not severely low (6.9–7.3 mg/100 ml) in our patient. Such a level is sometimes observed in hemodialysis patients who are not on treatment with calcium or vitamin D. However, the serum calcium level declined from 9.9 to 6.9 mg/100 ml within 2 days. In the previously reported cases (table 2), all but 2 (cases 6 and 8) had a lower serum calcium than our patient, and patients with profound hypocalcemia developed heart failure within several days (cases 9 and 11). In

spite of acute hypocalcemia, EKGs in this period did not show a prolonged Q-T interval so markedly as it was seen in a previously reported case [3]. The reason is not clear, but the mildness of hypocalcemia and/or the coexistence of hyperkalemia may be responsible.

The acute postoperative decline of serum calcium in our patient is considered to be a typical feature of the 'hungry bone' syndrome [5]. In the same way as Malluche et al. [7] described, deferoxamine treatment prior to parathyroidectomy might have facilitated calcium deposition in the bones and decrease of serum calcium during the postoperative course in our patient.

The long-term hemodialysis-associated organic heart disease may be the other important contributing factor for the development of hypocalcemic heart failure in our patient. Our patient had a history of cardiac arrest and low normal blood pressure, although overt heart failure was not observed in the past several years. Because of the cardiac damage, our patient may be prone to develop hypocalcemic heart failure rather than tetany.

Participation of magnesium deficiency in the development of cardiac dysfunction was suggested in hypoparathyroidism [8] and cardiovascular diseases [9]. However, it is not likely that magnesium played a primary role in the development of heart failure in our patient, since hypotension during hemodialysis was improved by calcium infusion alone without magnesium administration.

Parathyroidectomy is often performed for the treatment of renal osteodystrophy in long-term hemodialysis patients, who are highly likely to have myocardial injury. So one must take notice of exacerbation of latent myocardial dysfunction after parathyroidectomy in such patients.

In conclusion, a long-term hemodialysis patient developed heart failure by relatively mild hypocalcemia in the postoperative course of parathyroidectomy. The acute decline of serum calcium might exacerbate latent heart failure in the patient.

References

- 1 Fisch C: Electrolytes and the heart: in Hurst JW, et al. (eds): *The Heart, Arteries and Veins* ed 6. New York, McGraw-Hill, 1986, pp 1466-1479.
- 2 Nixon JV, Mitchell JH, McPhaul JJ Jr, Henrich WL: Effect of hemodialysis on left ventricular function. Dissociation of changes in filling volume and in contractile state. *J Clin Invest* 1983;71:377-384.
- 3 Chaimovitz C, Abinader E, Benderly A, Better OS: Hypocalcemic hypotension. *JAMA* 1972; 222:86-87.
- 4 Wong C-H, Pun K-K, Cheng C-H, Lau C-P, Leung W-H, Chan M-K, Yeung DWC: Hypocalcemic heart failure in end-stage renal disease. *Am J Nephrol* 1990;10:167-170.
- 5 Falko JM, Bush CA, Tzagournis M, Thomas FB: Congestive heart failure complicating the hungry bone syndrome. *Am J Med Sci* 1976;271: 85-89.
- 6 Weill J, Bouchard R, Antebi L, Guedeney J, Perles C: La cardiomégalie liée à une hypocalcémie chronique. *Soc Méd Hôp Paris* 1967; 118:659-667.
- 7 Malluche HH, Smith AJ, Abreo K, Faugere M-C: The use of deferoxamine in the management of aluminum accumulation in bone in patients with renal failure. *N Engl J Med* 1984;311:140-144.
- 8 Giles TD, Iteld BJ, Rives KL: The cardiomyopathy of hypoparathyroidism. Another reversible form of heart muscle disease. *Chest* 1981; 79:225-229.
- 9 Burch GE, Giles TD: The importance of magnesium deficiency in cardiovascular disease. *Am Heart J* 1977;94:649-657.
- 10 Rose E: Hypoparathyroidism. *Clinics* 1943; 1:1179-1196.
- 11 Evans JA, Elliott FD: Multiple vitamin deficiencies including beriberi heart with congestive failure. *Lahey Clin Bull* 1945;4:173-181.
- 12 Jernigan JA, Sadusk JF Jr: Idiopathic hypoparathyroidism: Discussion and presentation of a case. *Stanford Med Bull* 1953;11:266-271.
- 13 Grieve S, Schamroth L: Idiopathic hypoparathyroidism. *South Afr Med J* 1955;29:232-234.
- 14 Schulman JL, Ratner H: Idiopathic hypoparathyroidism with bony demineralization and cardiac decompensation. *Pediatrics* 1955;16: 848-856.
- 15 Bronsky D, Dubin A, Waldstein SS, Kushner DS: Calcium and the electrocardiogram: The electrocardiographic manifestations of hypoparathyroidism. *Am J Cardiol* 1961;7:823-832.
- 16 Antebi L, Bouchard R, Guedeney J, Perles C, Weill J: Cardiomegaly due to chronic hypocalcemia. *Pediatrics* 1966;38:909-911.
- 17 Najjar SS, Nassif SI, Khazen AF: Hypocalcemia and congestive heart failure. *Leb Med J* 1967;20:69-77.
- 18 Aryanpur I, Farhoudi A, Zangeneh F: Congestive heart failure secondary to idiopathic hypoparathyroidism. *Am J Dis Child* 1974;127: 738-739.
- 19 Brenton DP, Gonzales J, Pollard AB: Hypocalcemic cardiac failure. *Postgrad Med J* 1978; 54:633-636.
- 20 Bashour T, Basha HS, Cheng TO: Hypocalcemic cardiomyopathy. *Chest* 1980;78:663-665.
- 21 Connor TB, Rosen BL, Blaustein MP, Applefeld MM, Doyle LA: Hypocalcemia precipitating congestive heart failure. *N Engl J Med* 1982; 307:869-872.
- 22 Levine SN, Rheams CN: Hypocalcemic heart failure. *Am J Med* 1985;78:1033-1035.
- 23 Rimailho A, Bouchard P, Schaison G, Richard C, Auzépy P: Improvement of cardiomyopathy by correction of serum calcium level. *Am Heart J* 1985;109:611-613.